POWER BI DATA CLEANING AND PREPARATION USING POWER QUERY

Introduction

Data cleaning and preparation are important steps in data analysis. Raw data may contain errors, duplicate records, missing values, or incorrect formats. Before creating reports and dashboards in Power BI, the data should be cleaned and organized properly.

In this project, Power Query in Power BI was used to import, clean, and transform datasets. Different tools and techniques such as data import, data type conversion, text formatting, column creation, conditional columns, merging, filtering, grouping, and data modeling were used to

Project Objective

The main objective of this project is to clean and prepare raw datasets using Power Query in Power BI for better analysis and reporting.

Objectives:

- Import datasets into Power BI
- Change data types correctly
- Format text data
- Create new columns
- Merge datasets
- Handle missing and duplicate values
- Sort and filter data
- Group and summarize data
- Create table relationships

Dataset Overview

This project contains three datasets:

1. List of Orders

Contains customer order details such as:

- Order ID
- Order Date
- Customer Name
- State
- City

2. Order Details

Contains sales transaction details such as:

- Order ID
- Amount
- Profit
- Quantity
- Category
- Sub-Category

3. Sales Target

Contains target details such as:

- Month of Order Date
- Category
- Target

Data Import

The datasets List of Orders.csv, Order Details.csv, and Sales Target.csv were imported into Power BI using the Get Data option.

After importing, the datasets were opened in the Power Query Editor using the Transform Data option for further data cleaning and transformation.

Row Limiting & Data Types

The Keep Top Rows option was used to restrict the List of Orders table to the first 500 rows.

The following data type changes were made:

Column Name

Data Type

Order Date

Date

Amount

Fixed Decimal Number

Target

Fixed Decimal Number

This helped maintain data accuracy and consistency.

Text Formatting

The Format → Proper Case option was used to standardize the Customer Name column.

This ensured that all customer names followed a consistent text format and improved readability.

Column Creation

The Merge Columns option was used to combine City and State into a new column called Location.

Example:

Chennai, Tamil Nadu

A Custom Column named Profit Margin

Conditional Column

The Conditional Column option was used to create a new column called Profit Status.

The following conditions were applied:

Profit < 0 → Loss

Profit = 0 → Break-Even

Profit > 0 → Profit

This helped classify transactions based on profit performance.

Merging Data

The Merge Queries option was used to combine the List of Orders and Order Details tables using Order ID.

The merged table was renamed as Orders Data.

This process combined customer and sales information into a single dataset.

10. Handling Missing Data & Duplicate Data

The datasets were checked for missing values and duplicate records.

Null values were identified and handled wherever necessary. Duplicate rows were also checked and removed to improve data quality and avoid repeated records.

Sorting & Filtering Data

The Sort Descending option was used to sort the Orders Data table by Order Date.

Filters were applied to analyze records from specific states such as Tamil Nadu.

This helped focus on specific data for analysis.

Grouping & Aggregating Data

The Group By option was used to summarize the data.

The following calculations were performed:

Count of Order ID

Average Profit by Category

Total Amount by Sub-Category

Total Target by Month of Order Date

This helped convert raw data into summarized information.

Data Modeling

The Manage Relationships option was used to create relationships between tables.

Relationships Created:

List of Orders ↔ Order Details using Order ID

Order Details ↔ Sales Target using Category

This ensured proper connection between tables for accurate analysis.

Conclusion

This project successfully demonstrated the process of data cleaning and preparation using Power Query in Power BI. Different tools such as Get Data, Transform Data, Keep Top Rows, Proper Case, Merge Columns, Custom Column, Conditional Column, Merge Queries, Group By, Sort & Filter, and Manage Relationships were used to prepare the data.

The final dataset became more organized, accurate, and ready for reporting and dashboard creation in Power BI.